深度学习的物理学

课程幻灯片中英对照

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52 页全套交付候选：独立审校修订版

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Physics of Deep Learning
Part I: Bayesian learning and phase transitions

Tudor Dimofte

Iliad Intensive (Sept. 2026)

I. Bayesian learning and phase transitions
II. Networks at large width
III. Scaling laws
IV. Closing reading: learning mechanics

第 1 页：深度学习的物理学

第一部分：贝叶斯学习与相变。Tudor Dimofte，Iliad Intensive（2026 年 9 月）。

1. 贝叶斯学习与相变；

2. 大宽度下的网络；

3. 标度律；

4. 结束阅读：学习的机制。

Questions about learning

How and when are **features** learned during training?

- How to quantify? How to even *define* features?
- How does this depend on architecture, # parameters, training data, ...
- How can we *steer* learning, emergent capabilities, etc.?

Physics-inspired approaches help identify **approximations** in which these questions can be studied, in a simple(r), quantitative, often analytic way.

The approximations typically involve **scaling** of some sort: sending various (hyper)parameters $\to \infty$.

第 2 页：关于学习的问题

训练过程中，特征如何以及何时被学习？

- 如何量化特征？甚至应如何定义特征？
- 这如何依赖于架构、参数数量、训练数据等？
- 我们如何引导学习、涌现能力等？

受物理学启发的方法有助于确定一些近似，使我们能够以更简单、定量且通常可解析的方式研究这些问题。此类近似通常涉及某种标度：令各种（超）参数趋于无穷大。

Structure of the day

This lecture: Bayesian learning and statistical mechanics. A common approximation scheme, with a particularly strong connection to physics. I'll mainly talk about physics, and ask you to apply the same ideas to learning in the exercises!

Part II: Behavior of neural networks at large values of hyperparameters (especially: large width).

Part III: Neural scaling laws – one of the simplest robust empirical observations that any theoretical model needs to be able to explain.

Part IV: (if there's time) Reading & discussing a recent survey on the science of deep learning.

第 3 页：一天的安排

本次讲座讨论贝叶斯学习与统计力学，这是一种与物理学联系尤其紧密的常见近似方案。我主要讨论物理，并请大家在习题中把相同思想应用于学习！

- 第二部分：超参数取大值时神经网络的行为，尤其是大宽度。
- 第三部分：神经网络标度律——任何理论模型都必须能够解释的最简单、稳健的经验观察之一。
- 第四部分（如果有时间）：阅读并讨论一篇近期关于深度学习科学的综述。

Statistical mechanics & statistical learning

We initialize networks randomly:

$$\theta \sim \pi \qquad (\theta = \text{parameters},\ \#\theta = N)\,.$$

Then we train with some version of gradient descent, roughly

$$\Delta\theta \approx -\eta\,\nabla_\theta L(\theta)\,.$$

A common approximation to GD is **Bayesian learning**

$$p_\beta(\theta) \;\propto\; e^{-\beta L(\theta)}\,\pi(\theta)\,.$$

- Interpret $e^{-\beta L(\theta)}$ as the *likelihood* of the training data, given the model at fixed θ.
- $\beta \approx$ how much training has been done.

第 4 页：统计力学与统计学习

我们随机初始化网络：

$$\theta \sim \pi \qquad (\theta = \text{参数},\ \#\theta = N).$$

然后用某种梯度下降进行训练，大致为 $\Delta\theta \approx -\eta\nabla_\theta L(\theta)$。对梯度下降的一种常见近似是贝叶斯学习：

$$p_\beta(\theta) \propto e^{-\beta L(\theta)}\pi(\theta).$$

- 把 $e^{-\beta L(\theta)}$ 解释为给定固定参数 θ 的模型时，训练数据的似然。
- β 大致表示训练已经进行了多少。

An interpretation of β

For n samples $(x_i, y_i)_{i=1}^n$:

$$L_n(\theta) = \sum_{i=1}^{n} \ell_i(\theta)\,, \qquad \ell_i(\theta) = \ell\big(f_\theta(x_i),\, y_i\big)\,,$$

$$L_n(\theta) = n \cdot \underbrace{\frac{1}{n}\sum_{i} \ell_i(\theta)}_{\text{finite as } n\to\infty} \;\approx\; n\,\bar{L}(\theta)\,, \qquad \bar{L}(\theta) = \mathbb{E}_{(x,y)\sim q}\,\ell\big(f_\theta(x), y\big)\,.$$

For large n:

$$p_\beta(\theta) \;\propto\; e^{-\,n\beta\,\bar{L}(\theta)}\,\pi(\theta)\,.$$

- The effective parameter is $\beta' = n\beta$: *more data = more training = larger β' = "colder."*
- We'll often write $e^{-\beta \bar{L}}\pi$; but remember $\beta \to \infty$ as $n \to \infty$.

第 5 页：β 的一种解释

对于 n 个样本 $(x_i, y_i)_{i=1}^n$，

$$L_n(\theta) = \sum_{i=1}^{n} \ell_i(\theta), \qquad \ell_i(\theta) = \ell(f_\theta(x_i), y_i),$$

$$L_n(\theta) = n \cdot \frac{1}{n}\sum_{i} \ell_i(\theta) \approx n\,\overline{L}(\theta), \qquad \overline{L}(\theta) = \mathbb{E}_{(x,y)\sim q}\ell(f_\theta(x), y),$$

其中平均损失在 $n \to \infty$ 时有限。当 n 很大时，

$$p_\beta(\theta) \propto e^{-n\beta\overline{L}(\theta)}\pi(\theta).$$

有效参数是 $\beta' = n\beta$：更多数据对应更多训练、更大的 β'，也就是“更冷”。我们常写 $e^{-\beta\overline{L}}\pi$，但要记住当 $n \to \infty$ 时 $\beta \to \infty$。

The analogy with thermal physics

Bayesian learning looks a lot like a **thermal** system, i.e. like near-equilibrium statistical mechanics:

$$\begin{aligned} \theta \ \ (\#\theta = N) &\leftrightarrow \text{microscopic state of a system} \\ &\quad \text{(velocities of } N \text{ gas particles; spins of } N \text{ electrons)} \\ L(\theta) &\leftrightarrow \text{energy } E(\theta) \ \ \text{(or Euclidean action } S(\theta)) \\ p(\theta) \propto e^{-\beta L(\theta)} &\leftrightarrow \text{Boltzmann distribution: state of a system} \\ &\quad \text{that's been equilibrated with a bath at } T = 1/\beta \end{aligned}$$

Note: to include the prior $\pi(\theta)$ in this setup, just redefine

$$\tilde{L} = L - \tfrac{1}{\beta}\log\pi \quad \text{s.t.} \quad e^{-\beta L(\theta)}\pi(\theta) = e^{-\beta\tilde{L}(\theta)}\,, \qquad \tilde{L}(\theta) \leftrightarrow E(\theta)\,.$$

The analogy opens up lots of useful **methodology**.

第 6 页：与热物理的类比

贝叶斯学习看起来很像热系统，即接近平衡的统计力学：

$$\begin{aligned} \theta\ (\#\theta = N) &\longleftrightarrow \text{系统微观态（如 } N \text{ 个气体粒子的速度、} N \text{ 个电子的自旋）}, \\ L(\theta) &\longleftrightarrow \text{能量 } E(\theta) \text{（或欧氏作用量 } S(\theta)\text{）}, \\ p(\theta) \propto e^{-\beta L(\theta)} &\longleftrightarrow \text{与温度 } T = 1/\beta \text{ 的热浴达到平衡的系统的 Boltzmann 分布.} \end{aligned}$$

为在该设定中纳入先验 $\pi(\theta)$，只需重新定义

$$\widetilde{L} = L - \frac{1}{\beta}\log\pi, \qquad e^{-\beta L(\theta)}\pi(\theta) = e^{-\beta\widetilde{L}(\theta)}, \qquad \widetilde{L}(\theta) \longleftrightarrow E(\theta).$$

这一类比提供了许多有用方法。

Terminology & common tricks

Partition function (of β & other hyperparameters):

$$Z(\beta) = \int d\theta\ e^{-\beta E(\theta)}\,, \qquad p_\beta(\theta) = \tfrac{1}{Z(\beta)}\, e^{-\beta E(\theta)}\,.$$

Suppose there are **observables** $\mathcal{O}_i(\theta)$ whose expectation values we want. Generalize with **sources** J:

$$Z(\beta, J) = \int d\theta\ e^{-\beta E(\theta) + \sum_i J^i \mathcal{O}_i(\theta)}\,.$$

Then

$$\mathbb{E}(\mathcal{O}_i) = \frac{\partial}{\partial J^i} \log Z\Big|_{J=0}\,, \quad \mathrm{Cov}(\mathcal{O}_i, \mathcal{O}_j) = \frac{\partial^2}{\partial J^i \partial J^j} \log Z\Big|_{J=0}\,,$$

and in general $\kappa^{(p)}(\mathcal{O}_{i_1}, \ldots, \mathcal{O}_{i_p}) = \partial_{J^{i_1}} \cdots \partial_{J^{i_p}} \log Z\big|_{J=0}$ is the p-th **cumulant**.

$F(\beta, J) := -\log Z(\beta, J)$ is the (sourced) **free energy**: the generating function of cumulants.

第 7 页：术语与常用技巧

配分函数（依赖 β 及其他超参数）为

$$Z(\beta) = \int d\theta\, e^{-\beta E(\theta)}, \qquad p_\beta(\theta) = \frac{1}{Z(\beta)} e^{-\beta E(\theta)}.$$

假设存在希望求其期望值的可观测量 $O_i(\theta)$，引入源 J 加以推广：

$$Z(\beta, J) = \int d\theta\, e^{-\beta E(\theta) + \sum_i J^i O_i(\theta)}.$$

于是

$$\mathbb{E}(O_i) = \frac{\partial}{\partial J^i} \log Z\Big|_{J=0}, \qquad \mathrm{Cov}(O_i, O_j) = \frac{\partial^2}{\partial J^i \partial J^j} \log Z\Big|_{J=0}.$$

一般地，$\kappa^{(p)}(O_{i_1}, \ldots, O_{i_p}) = \partial_{J^{i_1}} \cdots \partial_{J^{i_p}} \log Z|_{J=0}$ 是 p 阶累积量。$F(\beta, J) := -\log Z(\beta, J)$ 是含源自由能，也就是累积量的生成函数。[1]

[1] 译注：按上面的定义，直接对 $F = -\log Z$ 作源导数与对 $\log Z$ 作源导数相差负号，需注意所用生成函数约定。

Cumulants of the energy itself

If the only observable we cared about was $E(\theta)$, we could just use $Z(\beta)$ (no sources), $F(\beta) = -\log Z(\beta)$:

$$\mathbb{E}(E) = \partial_\beta F\,, \qquad \mathrm{Var}(E) = -\partial_\beta^2 F\,, \qquad \kappa^{(3)}(E) = \partial_\beta^3 F\,, \ \ldots$$

We'll see these derivatives of F again in a little bit.

The analogy with physics also opens up a bag of well-developed **approximations**. We'll illustrate two in a classic physics example:

- Mean-field approximation, to help connect microscopic (individual param.) and macroscopic (large-scale) observables.
- Simplifications at large N ($N = \#$ parameters here, but other limits are similar)

We'll look at the *Ising model*; see Tong's *Statistical Field Theory* notes for more on the awesome physics of this example.

第 8 页：能量本身的累积量

如果唯一关心的可观测量是 $E(\theta)$，就可以直接使用 $Z(\beta)$（无需源）以及 $F(\beta) = -\log Z(\beta)$：

$$\mathbb{E}(E) = \partial_\beta F, \qquad \mathrm{Var}(E) = -\partial_\beta^2 F, \qquad \kappa^{(3)}(E) = \partial_\beta^3 F, \ldots$$

稍后还会再次见到 F 的这些导数。与物理学的类比还带来一整套成熟的近似方法。我们将通过一个经典物理例子介绍两种方法：

- 平均场近似，帮助联系微观（单个参数）与宏观（大尺度）可观测量；
- 大 N 下的简化（这里 N 是参数数量，其他极限也类似）。

我们将研究 Ising 模型；关于这个例子中的精彩物理，更多内容可参见 Tong 的《统计场论》讲义。

Physics example: the Ising model

Λ: a graph with N vertices $\{i\}$ and undirected edges $\{\langle ij\rangle\}$.
At each vertex, place a "spin" $\theta_i \in \{\pm 1\}$: a system with 2^N states, specified by $\theta_1, \ldots, \theta_N$.

$$E(\theta) \;=\; -J \sum_{\text{edges } \langle ij\rangle} \theta_i\, \theta_j \;-\; B \sum_{\text{vxs } i} \theta_i$$

- J: coupling strength (alignment with neighbors)
- B: external magnetic field (alignment with the field)

An interesting observable: the **magnetization**

$$m(\theta) = \frac{1}{N} \sum_{i=1}^{N} \theta_i \;\in\; [-1, 1] \qquad \text{(average spin in a given state)}\,.$$

The simplest *macroscopic* quantity describing the system.

第 9 页：物理例子——Ising 模型

Λ 是一个有 N 个顶点 $\{i\}$ 和无向边 $\{\langle ij\rangle\}$ 的图。在每个顶点放一个 "自旋" $\theta_i \in \{\pm 1\}$，系统共有 2^N 个状态，由 $\theta_1, \ldots, \theta_N$ 指定：

$$E(\theta) = -J \sum_{\text{边}\langle ij\rangle} \theta_i \theta_j - B \sum_{\text{顶点}i} \theta_i.$$

J 为耦合强度，促使自旋与邻居对齐；B 为外磁场，促使自旋与场对齐。一个有趣的可观测量是磁化强度

$$m(\theta) = \frac{1}{N} \sum_{i=1}^{N} \theta_i \in [-1, 1],$$

即给定状态的平均自旋，也是描述系统最简单的宏观量。

Mean-field approximation

We'd like to **push forward** the distribution $p_\beta(\theta) \propto e^{-\beta E(\theta)}$ to a distribution $p_\beta(m)$ over the macroscopic m.

Key: the **mean-field approximation**. Suppose each spin θ_i only feels the *average* force of all other spins, not its particular neighbors — i.e. replace each spin's actual neighbor sum by d copies of the global average (d = average # of neighbors):

$$\sum_{\langle ij\rangle} \theta_i\theta_j \;=\; \frac{1}{2}\sum_i \theta_i \underbrace{\sum_{j\sim i}\theta_j}_{\approx\; d\,m} \;\approx\; \frac{d}{2}\,m\sum_i \theta_i$$

$$\Rightarrow \quad E(\theta) \;\approx\; -N\Big(\frac{dJ}{2}\,m^2 + Bm\Big)\,.$$

Turns out to hold at large N and large $d \gtrsim 4$. Assuming this,

$$Z(\beta,J,B) \;=\; \sum_{\{\theta\}} e^{-\beta E(\theta)} \;\approx\; \int_{-1}^{1} dm\; e^{N\beta\left[\frac{dJ}{2}m^2+Bm\right]} \underbrace{\Omega(m)}_{\#\{\theta:\; m(\theta)=m\}}$$

第 10 页：平均场近似

我们希望把分布 $p_\beta(\theta) \propto e^{-\beta E(\theta)}$ 推前为宏观变量 m 的分布 $p_\beta(m)$。关键是平均场近似：假设每个自旋 θ_i 只感受到所有其他自旋的平均作用，而不感受其具体邻居。即将每个自旋的真实邻居和替换为 d 份全局平均，其中 d 为平均邻居数：

$$\sum_{\langle ij\rangle}\theta_i\theta_j = \frac{1}{2}\sum_i \theta_i \sum_{j\sim i}\theta_j \approx \frac{d}{2} m \sum_i \theta_i, \qquad\qquad E(\theta) \approx -N\left(\frac{dJ}{2}m^2 + Bm\right).$$

原文称该近似在 N 很大、$d \gtrsim 4$ 很大时成立。假定如此，则

$$Z(\beta,J,B) = \sum_{\{\theta\}} e^{-\beta E(\theta)} \approx \int_{-1}^{1} dm\, e^{N\beta[\frac{dJ}{2}m^2+Bm]}\Omega(m),$$

其中 $\Omega(m) = \#\{\theta : m(\theta) = m\}$。[2]

[2]译注：$d \gtrsim 4$ 本身不是能保证一般图上平均场近似成立的充分条件；结果依赖图结构及所取极限。

Entropy, and the reduction to one variable

Counting states at fixed m: $\Omega(m) = \binom{N}{\frac{1+m}{2}N}$, and Stirling gives

$$\log \Omega(m) \underset{N\to\infty}{\approx} N \underbrace{\left[\log 2 - \tfrac{1}{2}(1-m)\log(1-m) - \tfrac{1}{2}(1+m)\log(1+m)\right]}_{h(m)}$$

$h(m) = \frac{1}{N}\log(\#\text{ available states})$ is the **entropy** (per-site) at magnetization m: the effective measure after switching variables.

$$Z(\beta, J, B) = \int_{-1}^{1} dm \underbrace{e^{N\beta\left[\frac{dJ}{2}m^2 + Bm + \beta^{-1}h(m)\right]}}_{=:\ e^{-N\beta\, s(m)}}$$

$s(m)$: effective **action**, or free-energy density at m.

We've reduced the problem from N parameters to **one**.

第 11 页：熵与单变量化简

在固定 m 下计数状态：$\Omega(m) = \binom{N}{\frac{1+m}{2}N}$。利用 Stirling 公式，

$$\log \Omega(m) \underset{N\to\infty}{\approx} N\left[\log 2 - \frac{1}{2}(1-m)\log(1-m) - \frac{1}{2}(1+m)\log(1+m)\right] =: Nh(m).$$

$h(m) = N^{-1}\log$(可用状态数) 是磁化强度为 m 时的逐格点熵，也是在变换变量后的有效测度：

$$Z(\beta, J, B) = \int_{-1}^{1} dm\, e^{N\beta[\frac{dJ}{2}m^2 + Bm + \beta^{-1}h(m)]} = \int_{-1}^{1} dm\, e^{-N\beta s(m)}.$$

$s(m)$ 是有效作用量，或固定 m 时的自由能密度。我们已将问题从 N 个参数化简为一个。

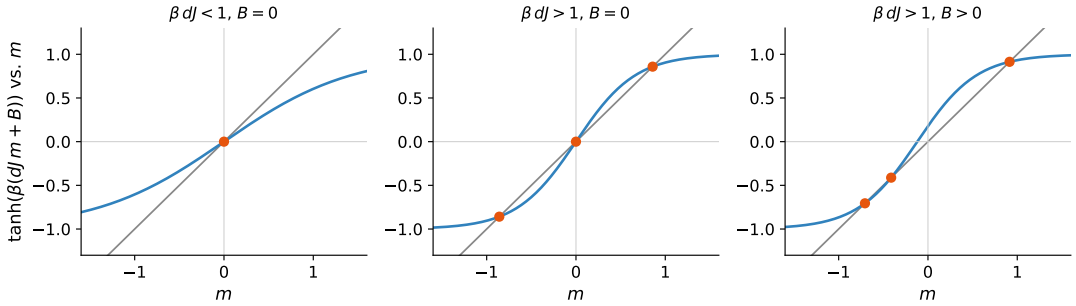
Saddle points

At large N, the integral concentrates around **minima** of s (saddle-point approximation):

$$\frac{\partial s}{\partial m} = 0 \quad \Longrightarrow \quad dJ\,\bar{m} + B = \beta^{-1}\tanh^{-1}(\bar{m})\,,$$

i.e. $$\boxed{\bar{m} = \tanh\left(\beta\left(dJ\,\bar{m} + B\right)\right)}$$

Solve graphically, for various β, J, B:



第 12 页：鞍点

在大 N 下，积分集中在 s 的极小值附近（鞍点近似）：

$$\frac{\partial s}{\partial m} = 0 \quad \Longrightarrow \quad dJ\bar{m} + B = \beta^{-1}\tanh^{-1}(\bar{m}),$$

即 $\bar{m} = \tanh\big(\beta(dJ\bar{m} + B)\big)$.

对不同的 β, J, B 作图求解。图中三栏分别标为 $dJ < 1, B = 0$；$dJ > 1, B = 0$；$dJ > 1, B > 0$，比较 $\tanh(\beta(dJm + B))$ 与 m。[3]

[3]译注：结合所写方程，图示分界一般应核查无量纲组合 βdJ；原图标签照录。

Self-consistency (an aside)

There's another path to the saddle-point equation. Look at the effective field/force felt by a single θ_i:

$$B^{\text{eff}} = -\partial_{\theta_i} E(\theta) = J \sum_{j \sim i} \theta_j + B \approx dJm + B\,.$$

The partition function for a single θ_i in this effective field becomes

$$Z_i = \sum_{\theta_i = \pm 1} e^{\beta B^{\text{eff}} \theta_i} = 2 \cosh \beta B^{\text{eff}} = 2 \cosh \beta (dJm + B)\,.$$

If the mean-field approx. is consistent, we'd better find

$$m \stackrel{!}{=} \mathbb{E}(\theta_i) = \tfrac{1}{\beta} \partial_B \log Z_i = \tanh \beta (dJm + B)$$

I.e. the average that comes *out* to equal the average we put in!

第 13 页：自洽性（补充）

还有另一种途径得到鞍点方程。考察单个 θ_i 感受到的有效场 / 力：

$$B^{\text{eff}} = -\partial_{\theta_i} E(\theta) = J \sum_{j \sim i} \theta_j + B \approx dJm + B.$$

在该有效场中，单个 θ_i 的配分函数为

$$Z_i = \sum_{\theta_i = \pm 1} e^{\beta B^{\text{eff}} \theta_i} = 2 \cosh(\beta B^{\text{eff}}) = 2 \cosh\big(\beta (dJm + B)\big).$$

如果平均场近似自洽，就应当有

$$m = \mathbb{E}(\theta_i) = \frac{1}{\beta} \partial_B \log Z_i = \tanh\big(\beta (dJm + B)\big),$$

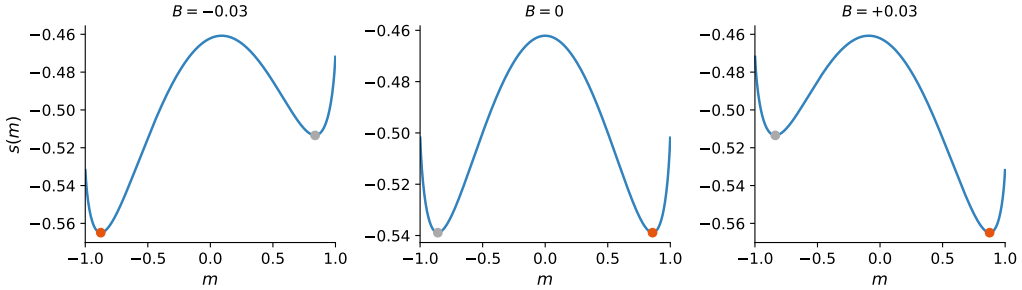
即计算得到的平均值等于我们投入的平均值！

Phases & phase transitions

Analyze the structure of **minima of $s(m)$** — they control where the probability density accumulates. Rich structure as β, J, B are varied! Taylor-expanding $h(m) \approx \log 2 - \frac{m^2}{2} - \frac{m^4}{12}$:

$$s(m) \;\approx\; \text{const} - B\,m + \tfrac{1}{2}\big(\beta^{-1} - Jd\big)\,m^2 + \tfrac{\beta^{-1}}{12}\,m^4 + \cdots$$

Take $Jd > \beta^{-1}$ (coupling beats temperature: the system tries to be *ordered*, neighboring spins align. But which way?)



They align **with the magnetic field B**: $\bar{m} \approx -1$ for $B < 0$, $\bar{m} \approx +1$ for $B > 0$. This is a **first-order phase transition** as B crosses $B_c = 0$.

第 14 页：相与相变

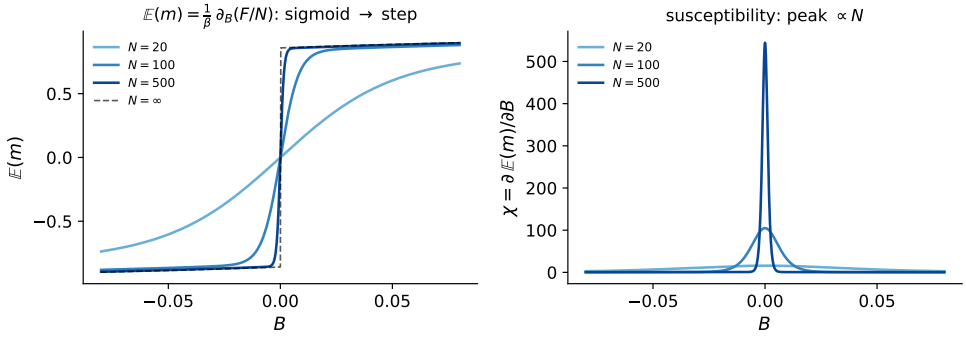
分析 $s(m)$ 的极小值结构，它们控制概率密度聚集的位置。改变 β, J, B 可得到丰富的结构。Taylor 展开 $h(m) \approx \log 2 - m^2/2 - m^4/12$，得到

$$s(m) \approx \text{const} - Bm + \frac{1}{2}(\beta^{-1} - Jd)m^2 + \frac{\beta^{-1}}{12}m^4 + \cdots .$$

取 $Jd > \beta^{-1}$（耦合胜过温度，系统倾向于有序，相邻自旋对齐；但朝哪个方向?）。图示分别为 $B = -0.03$、$B = 0$、$B = +0.03$。自旋沿磁场 B 对齐：$B < 0$ 时 $\bar{m} \approx -1$，$B > 0$ 时 $\bar{m} \approx +1$。当 B 越过 $B_c = 0$ 时，发生一级相变。

Features of a first-order transition

The structure of this transition, as two isolated minima cross, is universal (independent of the model). Look at $\mathbb{E}(m) = \frac{1}{N\beta}\frac{\partial F}{\partial B}$:



As $N \to \infty$, the saddle-point approx. strongly prefers m to sit in the lowest minimum, so $\mathbb{E}(m) = \beta^{-1}\partial_B(F/N)$ develops a discontinuity — the *definition* of a first-order phase transition. The "susceptibility" $\chi := \partial_B\mathbb{E}(m) = \beta^{-1}\partial_B^2(F/N)$ diverges to a δ.

第 15 页：一级相变的特征

两个孤立极小值交叉时的相变结构是普适的，即与具体模型无关。原文考察 $\mathbb{E}(m) = \frac{1}{N\beta}\partial_B F$。左图显示 $\mathbb{E}(m) = \beta^{-1}\partial_B(F/N)$ 从 sigmoid 趋向阶跃；右图显示磁化率 $\chi = \partial_B\mathbb{E}(m)$ 的峰随 N 增大。左图比较 $N = 20, 100, 500, \infty$，右图比较 $N = 20, 100, 500$；两图横轴均为 B。

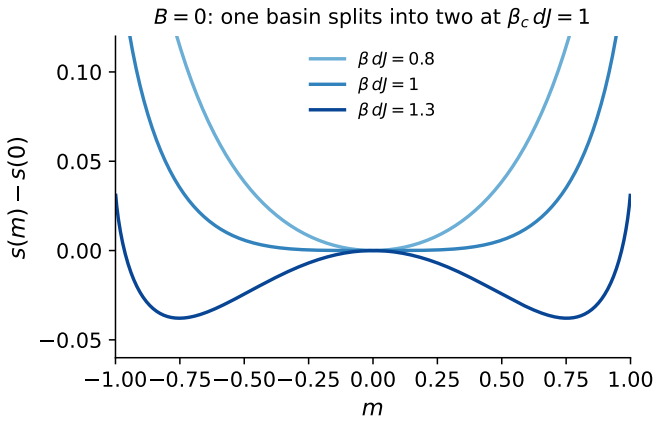
当 $N \to \infty$，鞍点近似强烈偏好 m 落在最低的极小值处，因此原文写出的 $\mathbb{E}(m) = \beta^{-1}\partial_B(F/N)$ 出现不连续——这就是一级相变的定义。原文还写"磁化率" $\chi := \partial_B\mathbb{E}(m) = \beta^{-1}\partial_B^2(F/N)$ 发散为 δ 函数。[4]

[4]译注：依第 7 页 $F = -\log Z$ 及第 9 页 $E = \cdots - B\sum_i \theta_i$ 的约定，$\mathbb{E}(m) = -\beta^{-1}\partial_B(F/N)$；因此本页两处自由能导数的正号需核查。

A second-order transition, too

An *n-th order* phase transition = discontinuity in the *n*-th derivative of F/N, as $N \to \infty$.

In the Ising model this happens at fixed $B = 0$ (symmetric minima), as the temperature $T = \beta^{-1}$ crosses $T_c = Jd$:



One basin deforms and splits: the order parameter $\mathbb{E}(m)$ moves *continuously* but not smoothly ($\mathbb{E}(m) \propto (\beta - \beta_c)^{1/2}$).

第 16 页：还有二级相变

n 级相变指当 $N \to \infty$ 时，F/N 的 n 阶导数不连续。在 Ising 模型中，固定 $B = 0$（对称极小值），当温度 $T = \beta^{-1}$ 越过 $T_c = Jd$ 时会发生这种相变。图中 $B = 0$，当 $\beta_c dJ = 1$ 时，一个势盆分裂为两个；示出 $\beta dJ = 0.8, 1, 1.3$ 三条曲线。一个势盆发生形变并分裂：序参量 $\mathbb{E}(m)$ 连续变化，但并不光滑，$\mathbb{E}(m) \propto (\beta - \beta_c)^{1/2}$。

Back to learning: Bayes vs. gradient descent

We don't train networks with Bayesian learning; we train them with **gradient descent**. Recall the expected analogy:

$$(*) \qquad \underbrace{n \to \infty}_{\text{samples / learning time}} \qquad \rightsquigarrow \quad \beta \to \infty \quad (T \to 0)\,.$$

- In Ising, this sweep would see the *second*-order transition.
- For *first*-order transitions, when we *dynamically* vary a parameter there is **hysteresis**: transitions are delayed. (One dynamical analogue in neural networks: the MSRJD formalism + DMFT — not in these lectures!)

第 17 页：回到学习——贝叶斯与梯度下降

我们不是用贝叶斯学习训练网络，而是用梯度下降。回忆期望中的类比：

$$\text{样本数 / 学习时间} \longrightarrow \infty \quad \rightsquigarrow \quad \beta \to \infty \quad (T \to 0).$$

- 在 Ising 模型中，这样的参数扫描会看到二级相变。
- 对一级相变，动态改变参数时存在滞后：相变发生得较晚。（神经网络中的一个动力学类比是 MSRJD 形式与 DMFT；本讲座不讨论！）

The Langevin equation (an aside)

Another connection to gradient descent, worth knowing: the **Langevin equation**

$$\dot{\theta} = -\nabla_\theta L(\theta) + T^{1/2}\, d\xi \qquad \text{(Gaussian noise, variance } T\text{)}.$$

- Starting from any sufficiently generic prior, at late training time ($t \to \infty$) this converges to $p(\theta) \propto e^{-\frac{1}{T}L(\theta)}$: one can "Bayesian train" a network in practice.
  (More on this in Tong's *Kinetic Theory* lectures.)
- Alternatively, $T \to 0$ at finite time t produces gradient flow $\dot{\theta} = -\nabla_\theta L$.

  $\rightsquigarrow$ a *different* Bayesian $\leftrightarrow$ gradient connection.

第 18 页：Langevin 方程（补充）

与梯度下降还有另一个值得了解的联系：Langevin 方程

$$\dot{\theta} = -\nabla_\theta L(\theta) + T^{1/2} d\xi \qquad \text{（高斯噪声，方差为 } T\text{）}.$$

- 从足够一般的先验出发，在训练时间很晚时（$t \to \infty$），它收敛到 $p(\theta) \propto e^{-L(\theta)/T}$：实践中可以对网络作“贝叶斯训练”。更多内容见 Tong 的《动力学理论》讲义。
- 另一种做法是在有限时间 t 取 $T \to 0$，得到梯度流 $\dot{\theta} = -\nabla_\theta L$，这给出了另一种贝叶斯与梯度下降之间的联系。

译注： 原页未明确白噪声 / Wiener 增量的归一化。如果将其理解为标准 Wiener 过程的 Itô 方程 $d\theta = -\nabla L\, dt + \sqrt{T}\, dW_t$，扩散系数为 $T/2$，相应平衡密度通常为 $e^{-2L/T}$；要得到所写的 $e^{-L/T}$，常用噪声幅度是 $\sqrt{2T}$。原式保留，具体因子需结合噪声约定核查。

Exercises (Part I)

See the handout. Do *one* of these.
Alternatively, go through formulas from the lecture and verify them.

0. Expectations and cumulants from the free energy.
1. **Anatomy of a first-order transition.** Work out the curves $\frac{\partial f}{\partial \beta}$ and $-\frac{\partial^2 f}{\partial \beta^2}$ $(f = F/N)$ for a generic first-order phase transition — and interpret them as the mean energy and the variance of the energy (heat capacity).
2. **The Ising perceptron.** Your turn! Analyze a simple learning problem and discover a first-order phase transition to perfect generalization.

第 19 页：习题（第一部分）

参见讲义，任选一道。也可以逐一检查本讲座的公式。

0. 从自由能求期望和累积量。
1. 一级相变的结构。对一般一级相变，计算 $\partial f/\partial \beta$ 与 $-\partial^2 f/\partial \beta^2$（$f = F/N$）的曲线，并将它们解释为平均能量和能量方差（热容）。
2. Ising 感知机。现在轮到你了！分析一个简单学习问题，发现通向完美泛化的一级相变。

References & further reading (Part I)

- D. Tong, *Statistical Field Theory* lecture notes — the physics behind this part (and *Kinetic Theory*, for the Langevin equation).
- A. Engel and C. Van den Broeck, *Statistical Mechanics of Learning* — the perceptron tradition; the Ising perceptron transition goes back to G. Györgyi (1990).
- The long lecture notes, *Physics of Deep Learning*, expand on everything today, with worked solutions.

第 20 页：参考文献与延伸阅读（第一部分）

- D. Tong，《统计场论》讲义：本部分背后的物理；Langevin 方程还可参见《动力学理论》。
- A. Engel 和 C. Van den Broeck，《学习的统计力学》：感知机研究传统；Ising 感知机相变可追溯至 G. Györgyi（1990）。
- 长篇讲义《深度学习的物理学》扩展了今天的全部内容，并附有详细解答。

译注：本地保存的长篇讲义是 nosol 版本，其附录 C 不含实际习题解答；幻灯片此处按原文翻译。

Physics of Deep Learning
Part II: Scaling neural networks and feature learning

Tudor Dimofte

Iliad Intensive (Sept. 2026)

Main question

Given a neural network, analyzed either dynamically (SGD) or via equilibrium Bayesian methods: *what scaling limits simplify the analysis?* (And are these limits realistic?)

1. Lazy scaling at initialization
2. Detour/reminder: the NTK & features
3. Lazy scaling during training
4. Mean-field scaling (and μP)

第 21 页：深度学习的物理学，第二部分

第二部分：神经网络标度与特征学习。Tudor Dimofte，Iliad Intensive（2026 年 9 月）。

主要问题：给定一个神经网络，无论是通过动力学（SGD）分析，还是通过平衡态贝叶斯方法分析，哪些标度极限能简化分析？这些极限是否现实？

1. 初始化时的惰性标度；
2. 补充 / 回顾：NTK 与特征；
3. 训练期间的惰性标度；
4. 平均场标度（及 μP）。

Setup: the MLP

A network with parameters $\theta \in \mathbb{R}^P$. Keep it simple: an **MLP** (fully-connected network), input $x \in \mathbb{R}^d$, L hidden layers each of width N, one output $f(x)$.

Pre-activations in layer ℓ:

$$z_i^{(\ell)} = \sum_{j=1}^{N} w_{ij}^{(\ell)}\, \phi\big(z_j^{(\ell-1)}\big)\,, \quad \text{except} \quad z_i^{(1)} = \sum_{j=1}^{d} w_{ij}^{(1)}\, x_j \;\; (\text{no } \phi)\,,$$

$$f = z^{(L+1)} \qquad \text{so} \qquad \theta = \big\{w^{(1)}, \ldots, w^{(L+1)}\big\}\,.$$

(No biases — they don't add anything conceptual.)

第 22 页：问题设定——多层感知机

考虑参数为 $\theta \in \mathbb{R}^P$ 的网络。为简单起见，取多层感知机（全连接网络）：输入 $x \in \mathbb{R}^d$，L 个隐藏层，每层宽度 N，一个输出 $f(x)$。第 ℓ 层的激活前量为

$$z_i^{(\ell)} = \sum_{j=1}^{N} w_{ij}^{(\ell)} \phi\big(z_j^{(\ell-1)}\big), \qquad \text{但} \quad z_i^{(1)} = \sum_{j=1}^{d} w_{ij}^{(1)} x_j \;\; (\text{不含 } \phi)\,,$$

$$f = z^{(L+1)}, \qquad \theta = \{w^{(1)}, \ldots, w^{(L+1)}\}.$$

这里没有偏置：它们不会增加新的概念。

Randomness at initialization

There is already randomness here, from random initialization: $\theta \sim \pi(\theta)$ (some *prior*).

Question: how does the randomness propagate to the values $f_\theta(x)$? Specifically: given the distribution π on θ's and fixed inputs $x^1, \ldots, x^k \in \mathbb{R}^d$, what is the **push-forward distribution** on

$$\big(f(x^1), \ldots, f(x^k)\big) \ ?$$

This simplifies greatly as $N \to \infty$.

But exactly *how* it simplifies depends strongly on π (as well as how learning rate, L, etc. scale with N).

第 23 页：初始化时的随机性

这里已经存在随机性，来自随机初始化：$\theta \sim \pi(\theta)$，即某个先验。

问题：随机性如何传播到函数值 $f_\theta(x)$？具体而言，给定 θ 的分布 π 与固定输入 $x^1, \ldots, x^k \in \mathbb{R}^d$，在下列量上的推前分布是什么？

$$\big(f(x^1), \ldots, f(x^k)\big).$$

当 $N \to \infty$ 时问题大大简化。但究竟如何简化，强烈依赖于 π，也依赖于学习率、L 等量如何随 N 标度。

Lazy initialization ("fan-in norm")

Lazy init: try to keep all pre-activations $z^{(\ell)} = O(1)$.

$$z_i^{(\ell)} = \sum_{j=1}^{N} \underbrace{w_{ij}^{(\ell)}}_{N \text{ random coeffts}} \underbrace{\phi\big(z_j^{(\ell-1)}\big)}_{O(1)}$$

$$\Longrightarrow \quad \text{take} \quad w_{ij}^{(\ell)} \sim \frac{1}{\sqrt{N}}\,, \qquad w_{ij}^{(1)} \sim \frac{1}{\sqrt{d}}\,.$$

Let's implement this explicitly: for $\ell > 1$ set

$$z_i^{(\ell)} = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} w_{ij}^{(\ell)}\, \phi\big(z_j^{(\ell-1)}\big)\,, \qquad w_{ij}^{(\ell)} \sim \mathcal{N}(0,1)\,,$$

so all the weights are now of order one.

第 24 页：惰性初始化（扇入归一化）

惰性初始化的目标是保持所有激活前量 $z^{(\ell)} = O(1)$。对于

$$z_i^{(\ell)} = \sum_{j=1}^{N} w_{ij}^{(\ell)} \phi\big(z_j^{(\ell-1)}\big),$$

其中有 N 个随机系数，而激活值为 $O(1)$，因此取

$$w_{ij}^{(\ell)} \sim \frac{1}{\sqrt{N}}, \qquad\qquad w_{ij}^{(1)} \sim \frac{1}{\sqrt{d}}.$$

显式实现这一点：当 $\ell > 1$，设

$$z_i^{(\ell)} = \frac{1}{\sqrt{N}} \sum_{j=1}^{N} w_{ij}^{(\ell)} \phi\big(z_j^{(\ell-1)}\big), \qquad\qquad w_{ij}^{(\ell)} \sim \mathcal{N}(0,1),$$

于是所有权重现在都是 $O(1)$。

Gaussian processes

Definition. Given $\mu : \mathbb{R}^d \to \mathbb{R}$ and $K : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ (the **kernel**), a **Gaussian process** is a random function $f : \mathbb{R}^d \to \mathbb{R}$ such that for any $(x^1, \dots, x^k)$, the joint distribution of $\big(f(x^1), \dots, f(x^k)\big)$ is Gaussian with mean $\big(\mu(x^1), \dots, \mu(x^k)\big)$ and covariance matrix

$$\begin{pmatrix} K(x^1, x^1) & \cdots & K(x^1, x^k) \\ \vdots & & \vdots \\ K(x^k, x^1) & \cdots & K(x^k, x^k) \end{pmatrix}.$$

Equivalently: mean and covariance exist, but *all higher cumulants vanish* (recall the cumulants of Part I — here the observables are the $f(x^i)$ themselves).

第 25 页：高斯过程

定义。给定 $\mu : \mathbb{R}^d \to \mathbb{R}$ 和 $K : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$（核），高斯过程是随机函数 $f : \mathbb{R}^d \to \mathbb{R}$，使得对任意 $(x^1, \dots, x^k)$，联合分布 $\big(f(x^1), \dots, f(x^k)\big)$ 都是高斯分布，均值为 $(\mu(x^1), \dots, \mu(x^k))$，协方差矩阵为

$$\begin{pmatrix} K(x^1, x^1) & \cdots & K(x^1, x^k) \\ \vdots & & \vdots \\ K(x^k, x^1) & \cdots & K(x^k, x^k) \end{pmatrix}.$$

等价地：均值与协方差存在，但所有更高阶累积量均消失（回忆第一部分的累积量；这里可观测量是 $f(x^i)$ 本身）。

Theorem: lazy init ⇒ Gaussian process

Theorem (Neal; Lee et al.; Yang; Hanin. . .). Fix L and x's. With lazy init, as $N \to \infty$, f converges to a Gaussian process, with recursively defined kernel:

$$K^{(\ell+1)}(x, x') = \mathbb{E}_{z \sim K^{(\ell)}}\Big[\phi\big(z(x)\big)\,\phi\big(z(x')\big)\Big],$$

with base case $K^{(1)}(x, x') = \frac{1}{d} x \cdot x'$. *Sketch:* the first layer is a sum of Gaussians — exactly Gaussian, with kernel $K^{(1)}$. Induct on layers: *conditioned* on layer ℓ, the $z_i^{(\ell+1)}$ are i.i.d. Gaussian with the *random* kernel $\widehat{K}^{(\ell)} = \frac{1}{N}\sum_j \phi(z_j^{(\ell)}(x))\,\phi(z_j^{(\ell)}(x'))$. Key: as $N \to \infty$ its variance $\to 0$, so it becomes *deterministic* — and Gaussianity propagates.

The recursion has an analytic form for $\phi = \mathrm{ReLU}$ and erf.

(There is also a slick physics derivation by path integrals — left as an exercise for those who like them; see the long notes, §3.)

第 26 页：定理——惰性初始化得到高斯过程

定理（Neal；Lee 等；Yang；Hanin 等）。固定 L 和各输入 x。在惰性初始化下，当 $N \to \infty$ 时，f 收敛到高斯过程，其核递归定义为

$$K^{(\ell+1)}(x, x') = \mathbb{E}_{z\sim K^{(\ell)}}[\phi(z(x))\phi(z(x'))], \qquad K^{(1)}(x, x') = \frac{1}{d} x \cdot x'.$$

证明概要：第一层是高斯变量之和，因此严格高斯，核为 $K^{(1)}$。对层数归纳：给定第 ℓ 层时，$z_i^{(\ell+1)}$ 独立同分布且高斯，随机核为

$$\widehat{K}^{(\ell)} = \frac{1}{N}\sum_j \phi(z_j^{(\ell)}(x))\phi(z_j^{(\ell)}(x')).$$

关键是当 $N \to \infty$ 时其方差趋于零，因此核成为确定量，而高斯性得以传播。$\phi = \mathrm{ReLU}$ 和 erf 时，递推关系有解析形式。（还可使用简洁的路径积分物理推导；留给有兴趣的读者作为习题，见长讲义第 3 节。）

Proof sketch, I: the first layer is exactly Gaussian

Fix the inputs $x^1, \dots, x^k$. The first layer, with the $1/\sqrt{d}$ explicit, is

$$z_i^{(1)}(x) = \frac{1}{\sqrt{d}} \sum_{j=1}^{d} w_{ij}^{(1)} x_j \,, \qquad w_{ij}^{(1)} \sim \mathcal{N}(0,1) \ \text{ i.i.d.}$$

For fixed i, this is a linear combination of independent Gaussians with *fixed* coefficients $x_j/\sqrt{d}$ — so $\big(z_i^{(1)}(x^1), \dots, z_i^{(1)}(x^k)\big)$ is **exactly jointly Gaussian** (no large-N limit needed), with mean zero and kernel

$$K^{(1)}(x,x') = \mathbb{E}\big[z_i^{(1)}(x)\, z_i^{(1)}(x')\big] = \frac{1}{d} \sum_{j,j'=1}^{d} x_j\, x'_{j'} \underbrace{\mathbb{E}\big[w_{ij}^{(1)} w_{ij'}^{(1)}\big]}_{\delta_{jj'}} = \frac{x \cdot x'}{d} \,.$$

Moreover, distinct neurons $i \neq i'$ use *disjoint* sets of weights, so the $z_i^{(1)}$ are i.i.d. across i: layer 1 is N independent copies of one Gaussian process.

第 27 页：证明概要 I——第一层严格高斯

固定输入 $x^1, \dots, x^k$。将 $1/\sqrt{d}$ 显式写出后，第一层为

$$z_i^{(1)}(x) = \frac{1}{\sqrt{d}} \sum_{j=1}^{d} w_{ij}^{(1)} x_j, \qquad\qquad w_{ij}^{(1)} \sim \mathcal{N}(0,1) \quad \text{独立同分布.}$$

固定 i 时，这是独立高斯变量的线性组合，系数 $x_j/\sqrt{d}$ 固定，因此 $(z_i^{(1)}(x^1), \dots, z_i^{(1)}(x^k))$ 严格联合高斯（不需要大 N 极限），均值为零，核为

$$K^{(1)}(x,x') = \mathbb{E}[z_i^{(1)}(x) z_i^{(1)}(x')] = \frac{1}{d} \sum_{j,j'=1}^{d} x_j x'_{j'}\, \mathbb{E}[w_{ij}^{(1)} w_{ij'}^{(1)}] = \frac{x \cdot x'}{d},$$

其中权重期望为 $\delta_{jj'}$。不同神经元 $i \neq i'$ 使用不相交的权重集合，所以 $z_i^{(1)}$ 在 i 上独立同分布：第一层是同一高斯过程的 N 个独立副本。

Proof sketch, II: induction on layers

Inductive hypothesis: as $N \to \infty$, the $z_j^{(\ell-1)}$ become i.i.d. GPs with deterministic kernel $K^{(\ell-1)}$.
Conditioned on layer $\ell-1$, the coefficients $\phi\big(z_j^{(\ell-1)}\big)$ are frozen, so $z_i^{(\ell)} = \frac{1}{\sqrt{N}} \sum_j w_{ij}^{(\ell)}\, \phi\big(z_j^{(\ell-1)}\big)$ is *again* linear in Gaussian weights: exactly Gaussian, i.i.d. across i, with kernel

$$\widehat{K}^{(\ell)}(x,x') = \frac{1}{N}\sum_{j=1}^{N} \phi\big(z_j^{(\ell-1)}(x)\big)\,\phi\big(z_j^{(\ell-1)}(x')\big)\,.$$

Un-conditionally, $\widehat{K}^{(\ell)}$ is a **random variable** — but an average of N i.i.d. $O(1)$ terms (by the inductive hypothesis), so $\mathrm{Var}\big[\widehat{K}^{(\ell)}\big] = O\big(\frac{1}{N}\big)$, with mean

$$\mathbb{E}\big[\widehat{K}^{(\ell)}\big] = \mathbb{E}_{z\sim K^{(\ell-1)}}\big[\phi\big(z(x)\big)\,\phi\big(z(x')\big)\big] = K^{(\ell)}(x,x')\,.$$

Key: as $N \to \infty$ the variance $\to 0$: the kernel becomes *deterministic*. A Gaussian mixture with deterministic kernel is Gaussian — and the neurons, coupled only through $\widehat{K}^{(\ell)}$, stay independent: Gaussianity propagates.

第 28 页：证明概要 II——按层归纳

归纳假设：当 $N \to \infty$，$z_j^{(\ell-1)}$ 成为具有确定核 $K^{(\ell-1)}$ 的独立同分布高斯过程。给定第 $\ell-1$ 层时，系数 $\phi(z_j^{(\ell-1)})$ 固定，故

$$z_i^{(\ell)} = \frac{1}{\sqrt{N}} \sum_j w_{ij}^{(\ell)} \phi(z_j^{(\ell-1)})$$

再次成为高斯权重的线性组合：它严格高斯，在 i 上独立同分布，其核为

$$\widehat{K}^{(\ell)}(x,x') = \frac{1}{N}\sum_{j=1}^{N} \phi(z_j^{(\ell-1)}(x))\phi(z_j^{(\ell-1)}(x')).$$

无条件地看，$\widehat{K}^{(\ell)}$ 是随机变量；原文称由归纳假设它是 N 个独立同分布 $O(1)$ 项的平均，因此

$$\mathrm{Var}[\widehat{K}^{(\ell)}] = O(1/N), \qquad \mathbb{E}[\widehat{K}^{(\ell)}] = \mathbb{E}_{z\sim K^{(\ell-1)}}[\phi(z(x))\phi(z(x'))] = K^{(\ell)}(x,x').$$

关键是当 $N \to \infty$ 时方差趋于零，核成为确定量。核确定的高斯混合仍为高斯，而仅通过 $\widehat{K}^{(\ell)}$ 耦合的神经元保持独立：高斯性得以传播。

Lesson — and the L/N expansion

Lesson: the $N \to \infty$ limit with lazy init is *boring* (but tractable!) at init. It also turns out boring(ish) in training.

But: the $N \to \infty$ limit does *not* commute with $L \to \infty$. If L is large, each successive layer gets larger fluctuations in its kernel, and they don't die off.

Hanin, Roberts–Yaida: for $N, L \to \infty$ networks the relevant expansion parameter is $\boxed{L/N}$. We've just analyzed $L/N = 0$.

- *Small* (but nonzero) L/N: no longer Gaussian — cumulants $K^{(4)}(f(x^1), \ldots, f(x^4))$ nonzero, but *controllable*.
- *Large* L/N: each successive cumulant is even larger; uncontrollable outputs.

This translates to **training** (SGD, GD, grad flow, even Bayesian):

$\frac{L}{N} = 0$: network is linear (features fixed);
$\frac{L}{N}$ small: features can be learned;
$\frac{L}{N}$ large: chaotic, divergent flows $\to$ untrainable network.

第 29 页：启示与 L/N 展开

启示：惰性初始化下的 $N \to \infty$ 极限在初始化时很无趣（但可处理!），训练时也相当无趣。

但是 $N \to \infty$ 极限与 $L \to \infty$ 不可交换。如果 L 很大，每一相继层的核涨落变得更大，而不再消失。Hanin、Roberts–Yaida：当 $N, L \to \infty$ 时，相关展开参数是 L/N；我们刚分析的是 $L/N = 0$。

- 小但非零的 L/N：不再高斯，累积量 $K^{(4)}(f(x^1), \ldots, f(x^4))$ 非零但可控制。
- 大 L/N：相继的累积量更大，输出不可控制。

这也适用于训练（SGD、GD、梯度流甚至贝叶斯）：$L/N = 0$ 时网络线性、特征固定；L/N 小时可以学习特征；L/N 大时流混沌或发散，网络不可训练。[5]

[5]译注：这一概括依赖具体架构、初始化、参数化和极限条件；不能仅由 L/N 推出对所有网络的无条件结论。

Detour: gradient flow and the NTK

To analyze training and quantify "features", we'll introduce a geometric object known as the Neural Tangent Kernel (NTK). (Jacot, Gabriel, Hongler 2018)

Recall: on a manifold X with smooth $h : X \to \mathbb{R}$ and a non-degenerate metric $g = g_{ab}(x)\, dx^a dx^b$, **gradient flow** is

$$\frac{dx^a}{dt} = -\, g^{ab}(x)\, \frac{\partial h}{\partial x^b} \qquad \text{(the flow of the vector field } -\, g^{-1}(dh)\text{)}.$$

You *need* a metric to define gradient flow, descent, etc.

In the parameter space of neural networks, the metric is assumed to be **Euclidean**: $g_\theta = \delta_{ij}\, d\theta^i d\theta^j$. (First source of inductive bias!)
Training (as gradient flow) is $\dot\theta^i = -\, \delta^{ij} \partial_j L(\theta)$.

第 30 页：补充——梯度流与 NTK

为了分析训练并量化 "特征"，我们引入称为神经切线核（NTK）的几何对象（Jacot、Gabriel、Hongler，2018）。

回忆：在带光滑函数 $h : X \to \mathbb{R}$ 和非退化度量 $g = g_{ab}(x) dx^a dx^b$ 的流形 X 上，梯度流为

$$\frac{dx^a}{dt} = -g^{ab}(x) \frac{\partial h}{\partial x^b} \qquad （向量场\ -g^{-1}(dh)\ 的流）.$$

定义梯度流、梯度下降等需要度量。在神经网络的参数空间中，假设度量为欧氏度量 $g_\theta = \delta_{ij} d\theta^i d\theta^j$。（归纳偏置的第一个来源！）作为梯度流的训练为 $\dot\theta^i = -\delta^{ij} \partial_j L(\theta)$。

Push the flow to function space: the NTK

Suppose $\{x^\mu\}_{\mu=1}^n$ are fixed inputs, with network outputs $f^\mu = f_\theta(x^\mu)$ and "truth" values y^μ. Push forward the flow from $\theta \in \mathbb{R}^P$ to $\{f^\mu\} \in \mathbb{R}^n$: the standard coordinate transformation of the *inverse metric* gives

$$\delta^{ij} \text{ on } \mathbb{R}^P \quad \rightsquigarrow \quad \Theta^{\mu\nu} = \frac{\partial f^\mu}{\partial \theta^i}\, \delta^{ij}\, \frac{\partial f^\nu}{\partial \theta^j} = \nabla_\theta f^\mu \cdot \nabla_\theta f^\nu \text{ on } \mathbb{R}^n\,,$$

$$\dot\theta = -\nabla_\theta L \quad \rightsquigarrow \quad \dot f^\mu = -\underbrace{\Theta^{\mu\nu}}_{\text{NTK}}\ \underbrace{\partial_{f^\nu} L}_{\text{residual}}\,.$$

Compactly, $\Theta = JJ^\top$ with $J = \nabla_\theta f$ the Jacobian.
For MSE loss

$$L = \frac{1}{2}\sum_\mu |f^\mu - y^\mu|^2 \quad \Rightarrow \quad \partial_{f^\mu} L = f^\mu - y^\mu =: \Delta^\mu\,.$$

"The NTK controls the geometry of optimization."

第 31 页：将流推前到函数空间——NTK

设 $\{x^\mu\}_{\mu=1}^n$ 是固定输入，网络输出为 $f^\mu = f_\theta(x^\mu)$，"真实" 值为 y^μ。将流从 $\theta \in \mathbb{R}^P$ 推前到 $\{f^\mu\} \in \mathbb{R}^n$；按原文逆度量的标准坐标变换，

$$\delta^{ij}(\text{在 } \mathbb{R}^P \text{ 上}) \quad \rightsquigarrow \quad \Theta^{\mu\nu} = \frac{\partial f^\mu}{\partial \theta^i}\delta^{ij}\frac{\partial f^\nu}{\partial \theta^j} = \nabla_\theta f^\mu \cdot \nabla_\theta f^\nu(\text{在 } \mathbb{R}^n \text{ 上})\,,$$
$$\dot\theta = -\nabla_\theta L \quad \rightsquigarrow \quad \dot f^\mu = -\Theta^{\mu\nu}\partial_{f^\nu} L.$$

这里 Θ 是 NTK，损失对 f 的导数是残差。简写为 $\Theta = JJ^T$，其中 $J = \nabla_\theta f$ 为 Jacobian。对均方误差损失，

$$L = \frac{1}{2}\sum_\mu |f^\mu - y^\mu|^2, \qquad \partial_{f^\mu} L = f^\mu - y^\mu =: \Delta^\mu.$$

"NTK 控制优化的几何。" [6]

[6]译注：JJ^T 可以退化；把它称为函数空间上的逆度量需核查秩和可投影等条件，而链式法则给出的流方程可直接计算。

Features, via the SVD

At a given θ, the Jacobian J says how the network values (on training data) respond to a shift in parameters. Take its SVD:

$$J = U \Lambda V^\top, \qquad \Lambda = \left(\operatorname{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots) \mid 0 \right).$$

- columns of V: modes in *parameter* space;
- columns of U: modes in *function values* $\rightarrow$ **"features"**;
- $\Theta = JJ^\top = U \Lambda \Lambda^\top U^\top$: the features are the **eigenvectors of the NTK**, with eigenvalues λ_k.

This is a commonly used theoretical notion of *feature*.

Cf. Activation-space or SAE features, grokked capabilities...

An empirical comparison: see J. Lin, *Feature Identification via the Empirical NTK*, arXiv:2510.00468: NTK eigenvectors matching features actually learned in modular arithmetic and a small language model.

第 32 页：通过 SVD 理解特征

在给定 θ 处，Jacobian J 描述训练数据上的网络函数值如何响应参数位移。取奇异值分解：

$$J = U\Lambda V^T, \qquad \Lambda = (\operatorname{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots) \;\; 0).$$

- V 的列：参数空间中的模式；
- U 的列：函数值中的模式，即“特征”；
- $\Theta = JJ^T = U\Lambda\Lambda^T U^T$：这些特征是 NTK 的特征向量，相应特征值为 λ_k。

这是一种常用的理论特征概念。可与激活空间或 SAE 特征、顿悟出的能力等比较。经验比较参见 J. Lin, *Feature Identification via the Empirical NTK*, arXiv:2510.00468：其中 NTK 特征向量与模算术及小语言模型中实际学到的特征相匹配。

How big is the NTK, in lazy scaling?

Back to training at large N! What does the NTK look like?

Try $L = 1$: $\quad f(x) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} c_i\, \phi(a_i \cdot x), \qquad a, c \sim \mathcal{N}(0,1).$

$$\nabla f = \frac{1}{\sqrt{N}} \bigoplus_i \Big(\phi(a_i \cdot x)\,,\ c_i\, \phi'(a_i \cdot x)\, x \Big)$$

$$\Theta^{\mu\nu} = \frac{1}{N} \sum_{i=1}^{N} \Big[\phi(a_i \cdot x^\mu)\, \phi(a_i \cdot x^\nu) + c_i^2\, \phi'(a_i \cdot x^\mu)\, \phi'(a_i \cdot x^\nu)\, x^\mu \cdot x^\nu \Big] = O(1)\,.$$

Similarly, at finite depth L and large N (with the $1/\sqrt{N}$'s explicit): Θ = sum of L terms, each $O(1)$ — so $\Theta = O(1)$, positive-definite.

Warning: in the $w \sim \mathcal{N}(0, \frac{1}{N})$ parametrization one finds $\Theta = O(N)$; the order-one flow there is $\dot\theta = -\frac{\eta}{N} \nabla L$. Rescaling parameters rescales the metric on parameter space — think of it as rescaling the *learning rate*. What matters: $O(1)$ changes in f^μ; figure out the size of Θ and scale η accordingly.

第 33 页：惰性标度下 NTK 有多大？

回到大 N 下的训练！NTK 是什么样的？先试 $L = 1$：

$$f(x) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} c_i \phi(a_i \cdot x), \qquad a, c \sim \mathcal{N}(0,1),$$

$$\nabla f = \frac{1}{\sqrt{N}} \bigoplus_i \big(\phi(a_i \cdot x),\ c_i \phi'(a_i \cdot x) x\big),$$

$$\Theta^{\mu\nu} = \frac{1}{N} \sum_{i=1}^{N} \left[\phi(a_i \cdot x^\mu)\phi(a_i \cdot x^\nu) + c_i^2 \phi'(a_i \cdot x^\mu)\phi'(a_i \cdot x^\nu) x^\mu \cdot x^\nu\right] = O(1).$$

类似地，在有限深度 L、大 N 且显式写出 $1/\sqrt{N}$ 时，Θ 是 L 个 $O(1)$ 项之和，故 $\Theta = O(1)$，正定。[7]

注意：在 $w \sim \mathcal{N}(0, 1/N)$ 的参数化中会得到 $\Theta = O(N)$，此时 $O(1)$ 的流为 $\dot\theta = -(\eta/N)\nabla L$。重标度参数会重标度参数空间的度量，可以把它看作重标度学习率。重要的是让 f^μ 出现 $O(1)$ 的变化；应先确定 Θ 的量级，再相应设置 η。

[7] 译注：JJ^T 一般只保证半正定；正定需要额外的秩条件。

Theorem: lazy training is lazy

Theorem (Du et al., Jacot et al., . . .). Fix depth, data, and a sufficiently overparametrized lazy network, trained by gradient flow with the scalings above. Despite the fact that $f^\mu = O(1)$ moves and fits the data:

- The network **linearizes** around initialization:

$$f_\theta(x) = f_{\theta_0}(x) + (\theta - \theta_0) \cdot \nabla_\theta f(x)\big|_{\theta_0} + O\Big(\frac{1}{\sqrt{N}}\Big);$$

- the NTK **freezes**: $\dfrac{\|\dot\Theta\|}{\|\Theta\|} = O\Big(\dfrac{1}{\sqrt{N}}\Big)$;
- $f(x^\mu) \to y^\mu$ *exponentially fast* in training time, with rates determined by the spectrum of Θ.

第 34 页：定理——惰性训练是惰性的

定理（Du 等；Jacot 等，等等）。固定深度和数据，取充分过参数化的惰性网络，以以上标度通过梯度流训练。虽然 $f^\mu = O(1)$ 会移动并拟合数据：

- 网络在初始化附近线性化：

$$f_\theta(x) = f_{\theta_0}(x) + (\theta - \theta_0) \cdot \nabla_\theta f(x)|_{\theta_0} + O(N^{-1/2});$$

- NTK 冻结：$\|\dot\Theta\|/\|\Theta\| = O(N^{-1/2})$；
- $f(x^\mu) \to y^\mu$，以由 Θ 的谱决定的速率随训练时间指数快收敛。

译注：这些结论需结合激活函数、训练轨迹、核的正定性和极限顺序等条件理解；若核有零模，不能无条件地断言所有残差模式都指数衰减到零。

Why? (sketch)

1. $\|\nabla^2 f\|_{\mathrm{op}} = O\big(\frac{1}{\sqrt{N}}\big)$: a big matrix, but *sparse* — block-diagonal per neuron, each block carrying the explicit $\frac{1}{\sqrt{N}}$. (In fact $O(\frac{1}{N})$ after cancelling fluctuations, but $\frac{1}{\sqrt{N}}$ is enough.)
2. Taylor-expand: the quadratic term is $\|\Delta\theta\|^2\,\|\nabla^2 f\|_{\mathrm{op}} \approx O(1) \times O\big(\frac{1}{\sqrt{N}}\big)$, using $\|\Delta\theta\| = O(1)$ from $\|\dot\theta\| = \|\sum_\mu \Delta^\mu \nabla f^\mu\| \sim \sqrt{\Theta} = O(1)$.
3. $\|\dot\Theta\| \le 2\,\|\nabla^2 f\|_{\mathrm{op}}\,\|\dot\theta\|\,\|\nabla f\| \to 0$: the kernel freezes.
4. With Θ constant, the flow solves itself: $\dot\Delta^\mu = -\Theta^{\mu\nu}\Delta_\nu$, so in the NTK eigenbasis ($\Theta = U\Lambda\Lambda^\top U^\top$)

$$(U^\top\Delta)_k(t) = e^{-\lambda_k t}\,(U^\top\Delta)_k(0)\ :$$

the k-th mode/feature is learned in time $\boxed{t_k \approx 1/\lambda_k}$.

第 35 页：为什么？(概要)

1. $\|\nabla^2 f\|_{\mathrm{op}} = O(N^{-1/2})$：这是一个大但稀疏的矩阵，按神经元分块对角，每块含显式的 $1/\sqrt{N}$。(抵消涨落后实际上是 $O(1/N)$，但 $1/\sqrt{N}$ 已足够。)
2. Taylor 展开：二次项 $\|\Delta\theta\|^2\|\nabla^2 f\|_{\mathrm{op}} \approx O(1)\times O(N^{-1/2})$；这里由 $\|\dot\theta\| = \|\sum_\mu \Delta^\mu\nabla f^\mu\| \sim \sqrt{\Theta} = O(1)$ 使用了 $\|\Delta\theta\| = O(1)$。
3. $\|\dot\Theta\| \le 2\|\nabla^2 f\|_{\mathrm{op}}\|\dot\theta\|\|\nabla f\| \to 0$，核因此冻结。
4. Θ 为常数时，流可直接求解：$\dot\Delta^\mu = -\Theta^{\mu\nu}\Delta^\nu$。因此在 NTK 特征基 $\Theta = U\Lambda\Lambda^T U^T$ 中，

$$(U^T\Delta)_k(t) = e^{-\lambda_k t}(U^T\Delta)_k(0),$$

第 k 个模式 / 特征的学习时间为 $t_k \approx 1/\lambda_k$。

译注：原页由速度量级直接推断整体位移量级的概要论证需要结合训练时间、残差衰减及一致界；不能仅凭速度为 $O(1)$ 就对任意长时间推出位移 $O(1)$。

Lazy learning = kernel regression

This is called **lazy learning**: it uses the (*random*!) features from initialization to approximate the actual features in the data.
Overparametrized system, so that's fine — but it only *memorizes*.

[The learning algorithm here is simple and well known: *kernel regression*.]

This is **not** a good model for actual neural networks, which *change* their NTK (phase transitions!) to adapt to the training data and generalize intelligently.
Remarkably, it's enough to predict some basic behavior (Part III).

Theoretical setups that do better? Yes! Simplest options:

- small $L/N > 0$ [see e.g. Roberts–Yaida–Hanin];
- a *different init*: we'll look at **mean-field scaling**

  [Generalized to μP (maximal-update param.), Yang & Hu 2019]

第 36 页：惰性学习等于核回归

这称为惰性学习：它使用初始化得到的（随机！）特征去逼近数据中的实际特征。系统是过参数化的，所以可以做到这一点——但它只是在记忆。（这里的学习算法简单且众所周知：核回归。）

这不是实际神经网络的好模型：实际网络会改变其 NTK（相变！），以适应训练数据并进行智能泛化。令人惊讶的是，它足以预测一些基本行为（第三部分）。更好的理论设定？有！最简单的选择：

- 小的 $L/N > 0$，例如见 Roberts–Yaida–Hanin；
- 不同的初始化：我们将研究平均场标度。Yang 和 Hu（2019）将其推广为 μP（最大更新参数化）。

译注：“只是在记忆”等是讲者的概括，核回归也可能具有泛化能力；其表现依赖核、数据和训练设定。

Mean-field scaling

For an $L = 1$ network, change $\frac{1}{\sqrt{N}} \to \frac{1}{N}$:

$$f(x) = \frac{1}{N} \sum_{i=1}^{N} c_i \, \phi(a_i \cdot x) \,, \qquad c_i, \, a_{ij} \sim \mathcal{N}(0,1) \,,$$

and send $N \to \infty$. Main differences from the lazy regime:

- At init, $f(x) \to 0$: the Gaussian process *collapses*.
- In training, $\Theta^{\mu\nu} = O\left(\frac{1}{N}\right)$ — which *forces* the learning rate to scale as N for anything to happen:

$$\dot{\theta} = -N \, \nabla_\theta L \,, \qquad \dot{f}^\mu = -N \, \Theta^{\mu\nu} \Delta^\nu \,.$$

$\rightsquigarrow$ a nonlinear (in parameters) network, with an **adaptive NTK**.
Why does the NTK move now?

$$\frac{|\dot{\Theta}|}{|\Theta|} \lesssim \underbrace{N}_{\text{l.r.}} \cdot \|\nabla^2 f\|_{\text{op}} = N \cdot O\left(\tfrac{1}{\sqrt{N}} \cdot \tfrac{1}{\sqrt{N}}\right) = O(1) \,,$$

so it *can* move; a closer analysis of $\dot{a}_i, \dot{c}_i$ shows that it *does*.

第 37 页：平均场标度

对 $L = 1$ 的网络，将 $1/\sqrt{N}$ 换成 $1/N$：

$$f(x) = \frac{1}{N} \sum_{i=1}^{N} c_i \phi(a_i \cdot x), \qquad c_i, a_{ij} \sim \mathcal{N}(0,1),$$

并令 $N \to \infty$。与惰性区域的主要区别：

- 初始化时 $f(x) \to 0$：高斯过程塌缩。
- 训练时 $\Theta^{\mu\nu} = O(1/N)$，若要发生非平凡变化，学习率必须按 N 标度：$\dot{\theta} = -N\nabla_\theta L$、$\dot{f}^\mu = -N\Theta^{\mu\nu}\Delta^\nu$。

由此得到对参数呈非线性的网络，其 NTK 可适应变化。为何 NTK 现在会移动？原文的量级估计为

$$\frac{|\dot{\Theta}|}{|\Theta|} \lesssim N \, \|\nabla^2 f\|_{\text{op}} = N \cdot O\left(\frac{1}{\sqrt{N}} \cdot \frac{1}{\sqrt{N}}\right) = O(1),$$

其中 N 来自学习率。因此它可以移动；更仔细分析 $\dot{a}_i, \dot{c}_i$ 可以看到它确实移动。

Why the name "mean field"?

Think of $(a_i, c_i)_{i=1}^N$ as parameters for N **particles**, each contributing $g_i^\mu := c_i\,\phi(a_i \cdot x^\mu)$ to $f(x^\mu)$. The loss/energy is

$$L = \frac{1}{2}\sum_{\mu=1}^{n}\big(f(x^\mu)-y^\mu\big)^2 = \frac{1}{N}\sum_{\mu=1}^{n}\left[\frac{1}{2N}\sum_{i,j=1}^{N} g_i^\mu\, g_j^\mu - y^\mu \sum_{i=1}^{N} g_i^\mu\right] + \text{const}$$

Same form as the mean-field approximation of Ising! Weak all-to-all couplings, depending only on the **averages**

$$G^\mu = \frac{1}{N}\sum_i g_i^\mu \;: \qquad L = \frac{1}{\cancel{N}}\cdot N \sum_\mu \left[\tfrac{1}{2}(G^\mu)^2 - y^\mu G^\mu\right].$$

Theorem. As $N \to \infty$, at all training times t, the distribution of particles converges,

$$\frac{1}{N}\sum_i \delta(a-a_i)\,\delta(c-c_i) \;\longrightarrow\; \mu_t(a,c)\;:$$

a *smooth, deterministic* measure obeying a deterministic PDE.

(Details, and a Bayesian attack on the same system: long notes, §5.)

第 38 页：为什么称为“平均场”？

将 $(a_i, c_i)_{i=1}^N$ 看成 N 个粒子的参数，每个粒子对 $f(x^\mu)$ 的贡献为 $g_i^\mu := c_i\phi(a_i \cdot x^\mu)$。损失 / 能量是

$$L = \frac{1}{2}\sum_{\mu=1}^{n}(f(x^\mu) - y^\mu)^2 = \frac{1}{N}\sum_{\mu=1}^{n}\left[\frac{1}{2N}\sum_{i,j=1}^{N} g_i^\mu g_j^\mu - y^\mu\sum_{i=1}^{N} g_i^\mu + \text{const}\right].$$

它具有与 Ising 平均场近似相同的形式：弱的全连接耦合只依赖平均值 $G^\mu = N^{-1}\sum_i g_i^\mu$，即

$$L = \frac{1}{N}\cdot N\sum_\mu\left[\frac{1}{2}(G^\mu)^2 - y^\mu G^\mu\right].$$

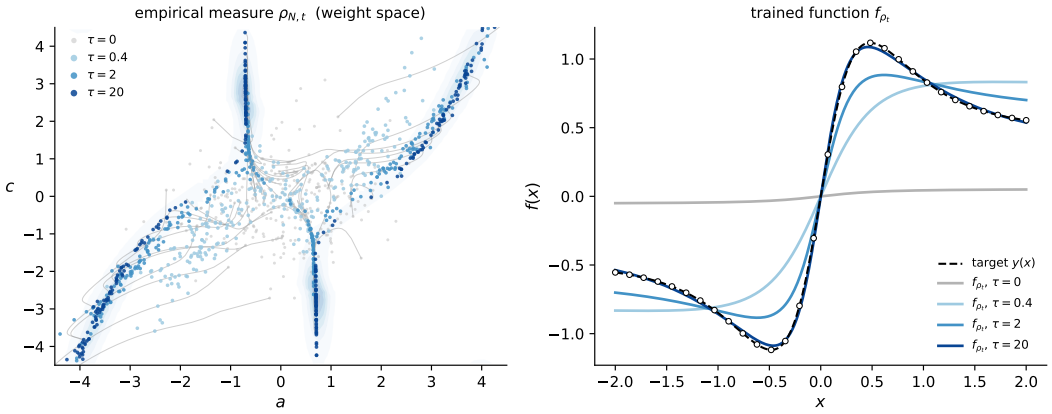
定理。当 $N \to \infty$，在所有训练时间 t，粒子分布收敛：

$$\frac{1}{N}\sum_i \delta(a-a_i)\delta(c-c_i) \longrightarrow \mu_t(a,c),$$

得到服从确定性偏微分方程的光滑、确定性测度。细节以及对同一系统的贝叶斯处理见长讲义第 5 节。[8]

[8] 译注：测度极限、光滑性和时间范围需依具体初始化、激活函数及动力学条件表述；此处保留幻灯片概述。

Feature learning, watched directly



Training in mean-field scaling transports the *cloud of neurons* (a_i, c_i) (left: dots colored by time, over the large-N density), which condenses onto the low-dimensional structures realizing the target — learning its features — while the network function converges (right). This is the model of Exercise 2.

第 39 页：直接观察特征学习

在平均场标度下训练会输运神经元云 (a_i, c_i)：左图中点的颜色表示时间，背景为大 N 密度；神经元云凝聚到实现目标的低维结构上，即学习其特征，同时网络函数收敛（右图）。这是习题 2 的模型。

左图横轴 a、纵轴 c，表示权重空间中的经验测度 $\mu_{N,t}$；图例给出的重标度时间为 $\tau = 0, 0.4, 2, 20$。右图横轴 x、纵轴 $f(x)$，比较目标 $y(x)$ 和同样四个时刻的训练函数 f_t。

Exercises (Part II)

See the handout:

0. **Depth-to-width ratios in the wild.** Look up L/N for open-weight LLM families: is everyone training in the "controlled correlations" window?
1. **Bayesian lazy learning.** The Bayesian-learning version of the lazy story: eigenmodes of the GP kernel at init are learned with characteristic "time" $\beta_k = 1/\lambda_k$ — the equilibrium twin of $t_k = 1/(\eta\lambda_k)$.
2. **Numerics: lazy vs. mean-field.** Train one small network in both scalings. Watch the NTK freeze in one and move in the other, and watch the particle cloud (a_i, c_i) condense onto features.

第 40 页：习题（第二部分）

参见讲义：

0. 实际模型中的深宽比。查阅开放权重 LLM 家族的 L/N：大家是否都在“受控关联”窗口内训练?
1. 贝叶斯惰性学习。惰性机制的贝叶斯学习版本：初始化 GP 核的特征模式以特征“时间”$\beta_k = 1/\lambda_k$ 被学习，它是 $t_k = 1/(\eta\lambda_k)$ 的平衡态对应物。
2. 数值实验：惰性与平均场。在两种标度下训练一个小网络，观察 NTK 在一种情况下冻结、在另一种情况下移动，并观察粒子云 (a_i, c_i) 凝聚成特征。

References & further reading (Part II)

- B. Hanin, *Mathematics of Deep Learning* lectures — large N `https://www.youtube.com/watch?v=4SVWo5kspR8`
- D. Roberts, S. Yaida, and B. Hanin, *The Principles of Deep Learning Theory* — the L/N expansion.
- Z. Ringel, N. Rubin, E. Mor, M. Helias, and I. Seroussi, *Applications of Statistical Field Theory in Deep Learning*.
- K. Spiliopoulos, R. Sowers, and J. Sirignano, *Mathematical Foundations of Deep Learning Models and Algorithms* — mean-field limits.
- The long lecture notes, *Physics of Deep Learning*, §3–§5, with worked solutions.

第 41 页：参考文献与延伸阅读（第二部分）

- B. Hanin，*Mathematics of Deep Learning* 讲座：大 N；`https://www.youtube.com/watch?v=4SVWo5kspR8`。
- D. Roberts、S. Yaida 和 B. Hanin，*The Principles of Deep Learning Theory*：L/N 展开。
- Z. Ringel、N. Rubin、E. Mor、M. Helias 和 I. Seroussi，*Applications of Statistical Field Theory in Deep Learning*。
- K. Spiliopoulos、R. Sowers 和 J. Sirignano，*Mathematical Foundations of Deep Learning Models and Algorithms*：平均场极限。
- 长讲义《深度学习的物理学》第 3–5 节，附详细解答。

译注：本地长讲义为 nosol 版本，未含实际习题解答；这里保留幻灯原文说明。

Physics of Deep Learning
Part III: Neural scaling laws

Tudor Dimofte

Iliad Intensive (Sept. 2026)

Organizing question

The test loss of large models falls as a clean power law in every resource spent on it. *Where do the exponents come from?*

第 42 页：深度学习的物理学，第三部分

第三部分：神经网络标度律。Tudor Dimofte，Iliad Intensive（2026 年 9 月）。

贯穿问题：大型模型的测试损失随投入的各种资源呈整洁的幂律下降。这些指数从何而来？

The empirical laws (Kaplan et al. 2020)

For autoregressive language models, when not bottlenecked by the other resources, test loss obeys

$$L(N) \propto N^{-\alpha_N}, \qquad L(D) \propto D^{-\alpha_D}, \qquad L(C) \propto C^{-\alpha_C},$$

with

- N = parameters (non-embedding)
- D = dataset size (in tokens)
- C = training compute (in FLOPs).

(Note: for single-epoch training, $C \approx 6ND$, with N & D optimally allocated.)

Measured exponents, holding over many orders of magnitude:

$$\alpha_N \approx 0.076, \qquad \alpha_D \approx 0.095, \qquad \alpha_C \approx 0.050.$$

第 43 页：经验定律（Kaplan 等，2020）

对于自回归语言模型，当其他资源不构成瓶颈时，测试损失服从

$$L(N) \propto N^{-\alpha_N}, \qquad L(D) \propto D^{-\alpha_D}, \qquad L(C) \propto C^{-\alpha_C},$$

其中 N 是非嵌入参数数量，D 是数据集大小（token 数），C 是训练计算量（FLOPs）。对单轮训练，$C \approx 6ND$，其中 N, D 作最优分配。测得的指数在许多数量级上成立：

$$\alpha_N \approx 0.076, \qquad \alpha_D \approx 0.095, \qquad \alpha_C \approx 0.050.$$

Universal form; data-dependent exponents

The same power-law *form* holds across modalities (Henighan et al. 2020) — but the *exponents* follow the data:

modality	α_N	α_C
language	0.070	0.048
images 32×32	0.13	0.10
images 8×8	0.24	0.19
video	0.24	0.14
math (procedural)	0.16	0.17

- **Data matters**: exponents vary by a factor of ∼3–5 across modalities.
- **Architecture barely matters**: within reasonable architectures, changing shape (depth vs. width) or family moves constants and plateaus, not exponents. Whatever sets the exponent lives in the *data*.

第 44 页：普适形式与依赖数据的指数

相同的幂律形式出现在不同模态中（Henighan 等，2020），但指数随数据变化：

模态	α_N	α_C
语言	0.070	0.048
32×32 图像	0.13	0.10
8×8 图像	0.24	0.19
视频	0.24	0.14
数学（程序生成）	0.16	0.17

- 数据重要：不同模态的指数变化约 3–5 倍。
- 架构几乎不重要：在合理的架构范围内，改变形状（深度与宽度）或家族，会改变常数和平台，而不改变指数。决定指数的因素存在于数据中。

译注：上述是幻灯片对所讨论实验的概括；它本身不证明所有架构或数据设置下指数均具有普适性。

Joint laws and bottlenecks

Single-resource laws assume the *other* resources are unlimited. Joint fits describe the crossover to being **bottlenecked**: loss is governed by whichever resource is scarcest. Two proposals:

Kaplan et al.: $L(N, D) = \left[\left(\frac{N_c}{N}\right)^{\alpha_N/\alpha_D} + \frac{D_c}{D}\right]^{\alpha_D}$

Chinchilla (Hoffmann et al. 2022): $L(N, D) = E + \frac{N_c}{N^{\alpha_N}} + \frac{D_c}{D^{\alpha_D}}$

- E: data's intrinsic noise floor, irreducible entropy (≈ 1.69 for text)
- Both forms reduce to single-resource power laws as either N or $D \to \infty$, removing bottleneck.
- The difference matters for compute-optimal scaling: at fixed $C \approx 6ND$, how to balance N and D?

第 45 页：联合规律与瓶颈

单资源规律假设其他资源无限。联合拟合描述进入瓶颈的交叉过程：损失由最稀缺的资源支配。两种方案为

$$\text{Kaplan 等：} \quad L(N, D) = \left[\left(\frac{N_c}{N}\right)^{\alpha_N/\alpha_D} + \frac{D_c}{D}\right]^{\alpha_D},$$

$$\text{Chinchilla（Hoffmann 等，2022）：} \quad L(N, D) = E + \frac{N_c}{N^{\alpha_N}} + \frac{D_c}{D^{\alpha_D}}.$$

- E 为数据的内禀噪声底、不可约熵（文本中约为 1.69）。
- 两种形式在 N 或 D 任一趋于无穷、移除瓶颈时都退化为单资源幂律。
- 这种差异会影响计算最优标度：固定 $C \approx 6ND$ 时，应如何平衡 N 和 D?

Compute-optimal scaling: Kaplan vs. Chinchilla

Kaplan et al. 2020	$N \propto C^{0.73}$, $D \propto C^{0.27}$: *grow the model*
Chinchilla 2022	$N \propto C^{0.5}$, $D \propto C^{0.5}$: *grow both equally*
	(rule of thumb: $\sim$ 20 tokens per parameter)

Which is right? Chinchilla's is now the accepted baseline. The discrepancy was analyzed & understood by Porian et al. 2024.

Note: In practice, frontier models now *over*-train relative to Chinchilla (more data per parameter), because smaller models are cheaper at *inference* time.

Name: *Chinchilla* was DeepMind's 70B model — trained with 4$\times$ the data of its 280B predecessor *Gopher* at the same compute, and beating it.

第 46 页：计算最优标度——Kaplan 与 Chinchilla

Kaplan 等（2020）：	$N \propto C^{0.73}$,	$D \propto C^{0.27}$（增大模型），
Chinchilla（2022）：	$N \propto C^{0.5}$,	$D \propto C^{0.5}$（两者同等增长）.

经验法则是每个参数约 20 个 token。哪个正确？原文称 Chinchilla 现为公认基线，Porian 等（2024）分析并解释了差异。注意：实践中前沿模型现在相对于 Chinchilla 进行过量训练（每个参数对应更多数据），因为较小模型在推理时更便宜。

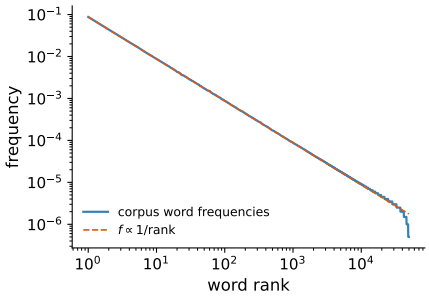
名称说明：Chinchilla 是 DeepMind 的 700 亿参数模型；在与其 2800 亿参数前代 Gopher 相同的计算量下，它使用四倍数据进行训练并取得更好的结果。

Where do the laws come from?

Basic answer: the loss curve is reading off the **coarse distribution of structure in the data**: power laws in, power laws out.

Simplest example: **Zipf's law**. Rank the words of a corpus by frequency, $f(\text{rank } k) \propto 1/k$.

(Similar: fact mentions, n-grams, image features ... $\propto k^{-\alpha}$)



Theoretical status: many models derive the *single-resource* laws from data structure (now fairly clear); a few derive *multi-resource* laws. Still missing: the data's exponents from first principles; joint laws with full feature-learning dynamics; extending to emergent *capabilities*. Also missing theory for empirical RL laws (beyond this lecture).

第 47 页：这些规律从哪里来？

基本回答：损失曲线读出了数据中结构的粗粒度分布——输入是幂律，输出也是幂律。

最简单的例子是 Zipf 定律：按频率排列语料中的词，$f(\text{排名}k) \propto 1/k$。类似地，事实提及、$n$-gram、图像特征等可服从 $k^{-\alpha}$。图中横轴为词的排名（word rank），纵轴为词频（frequency），均为对数刻度；图例“corpus word frequencies”表示语料词频，“$f \propto 1/\text{rank}$”表示反比参考线。

理论现状：许多模型由数据结构导出单资源规律，少数导出多资源规律。仍然缺少的是从第一性原理得到数据指数、具有完整特征学习动力学的联合规律、向涌现能力的推广。关于经验强化学习规律的理论也仍然缺失（超出本讲座范围）。

A survey of theoretical proposals

proposal	data model / "features"	NN framework	exponents?	multi-param?
Sharma–Kaplan 2020	smooth target on d-dim. manifold	piecewise-linear interpolation	yes: $\alpha_N \approx 4/d$	no (N only)
kernel eigenlearning; Bahri et al. 2021; Bordelon et al.; Sollich (GP)	power-law kernel spectra + target coefficients	frozen kernel: NTK / GP-Bayesian	in terms of spectral exponents	partially: resolution vs. variance
Maloney–Roberts–Sully 2022	latent power-law covariance, extended by features	random feature model (frozen NTK) at $t \to \infty$	in terms of data spectral exp.	yes: joint $L(N, D)$; duality $N \leftrightarrow D$
quantization; Michaud et al. 2023	discrete skill "quanta," Zipf-used, learned in order	none (counting argument)	in terms of Zipf α; relation $\frac{1}{\alpha_D} = \frac{1}{\alpha_N} + 1$	separate N, D, C laws
percolation; Brill 2024	subtask clusters at criticality (power-law sizes)	general learner + regression estimates	from percolation critical exponents	two regimes; not joint
DMFT; Bordelon–Atanasov–Pehlevan 2024	power-law (source/capacity) spectra	DMFT of gradient-flow training, w/ frozen NTK	yes, incl. time exponent	fully joint $L(t, N, D)$ + compute frontier

第 48 页：理论方案综览

原表的五列及六个方案如下。这里沿用幻灯片的参数数目 N、数据量 D 记号。

方案	数据模型 / "特征"	神经网络框架	是否给出指数	是否涉及多参数
Sharma–Kaplan（2020）	d 维流形上的光滑目标	分段线性插值	是：$\alpha_N \approx 4/d$	否（只有 N）
核本征学习；Bahri 等（2021）；Bordelon 等；Sollich（GP）	幂律核谱与目标系数	冻结核：NTK / GP 贝叶斯	由谱指数表示	部分处理：分辨率与方差
Maloney–Roberts–Sully（2022）	潜在幂律协方差，经特征扩展	$t \to \infty$ 的随机特征模型（冻结 NTK）	由数据谱指数表示	是：联合 $L(N, D)$；$N \leftrightarrow D$ 对偶
量化；Michaud 等（2023）	离散技能"量子"，按 Zipf 规律使用、依次学习	无（计数论证）	由 Zipf 指数 α 表示；$1/\alpha_D = 1/\alpha_N + 1$	分别给出 N, D, C 规律
渗流；Brill（2024）	临界时的子任务簇（大小服从幂律）	一般学习器与回归估计	来自渗流临界指数	两个区域；非联合规律
DMFT；Bordelon–Atanasov–Pehlevan（2024）	幂律的源 / 容量谱	梯度流训练的 DMFT；冻结 NTK	是，包括时间指数	完全联合 $L(t, N, D)$ 及计算前沿

Reading the table

- **Everyone agrees on the origin**: exponents are inherited from a power-law tail *in the data*. The models differ in what the tail is a tail *of*, and in how much network machinery is retained.
- Bayesian/equilibrium routes (GP learning curves; Ringel et al.'s effective-kernel methods at finite width) and dynamical routes (DMFT) give matching exponents where they overlap. *Cf.* Part I "two limits of one system".
- Only the static (MRS) and dynamical (DMFT) field-theory models currently predict *joint* laws — and DMFT derives a Chinchilla-like compute-optimal frontier. But both of these work in a frozen-NTK regime.
- None of them computes the *data's* exponent from first principles. Theory converts data statistics into loss curves; the statistics themselves are an empirical input.

第 49 页：解读表格

- 原文称大家都同意指数源于数据中的幂律尾；模型的区别在于尾部对应什么，以及保留多少网络机制。
- 贝叶斯 / 平衡态路径（GP 学习曲线、Ringel 等的有限宽度有效核方法）与动力学路径（DMFT）在重合的范围给出匹配的指数。可与第一部分"一个系统的两个极限"比较。
- 原文称目前仅静态（MRS）和动力学（DMFT）场论模型预测联合规律，DMFT 还导出类似 Chinchilla 的计算最优前沿；但两者均工作于冻结 NTK 区域。
- 它们都没有从第一性原理计算数据指数。理论把数据统计转换成损失曲线，而统计本身是经验输入。

译注：本页是讲者对所列模型的概括，未在此独立核验每项文献或排除其他理论方案。

The quantization model (Michaud et al. 2023)

A minimal model: no architecture, no optimizer, one tail sum.
Three assumptions:

- **QH1 (discreteness).** What a network learns decomposes into enumerable **quanta** — basic skills or facts ("carry the one"). Each is fully learned or not at all.
- **QH2 (ordered acquisition).** Models learn quanta in order of decreasing usefulness: a given model knows the first m.
- **QH3 (Zipfian use).** Quantum k is *used* on a fraction $p_k \propto k^{-(\alpha+1)}$ of data samples, and knowing it reduces the loss there by a constant Δ.

So a model that has learned m quanta has expected loss

$$L_m = L_\infty + \Delta \sum_{k>m} p_k \,.$$

The analysis boils down to asking **how each resource limits** m.

第 50 页：量化模型（Michaud 等，2023）

这是一个极简模型：没有架构、没有优化器，只有一个尾和。三项假设：

- QH1（离散性）：网络学到的东西可分解成可枚举的量子，即基本技能或事实（例如 "逢十进一"）；每项要么完全学会，要么完全不会。
- QH2（有序获取）：模型按用途递减的顺序学习量子；给定模型知道前 m 个。
- QH3（Zipf 使用规律）：第 k 个量子用于比例 $p_k \propto k^{-(\alpha+1)}$ 的数据样本；知道它能使相应损失减少常数 Δ。

因此，学会 m 个量子的模型具有期望损失

$$L_m = L_\infty + \Delta \sum_{k>m} p_k.$$

分析最终归结为：每种资源如何限制 m。

One law, worked; two laws, yours

Tail sum: $L_m - L_\infty = \Delta \sum_{k>m} p_k \;\propto\; \int_m^\infty k^{-(\alpha+1)}\,dk \;\propto\; m^{-\alpha}\,.$

Parameter-limited: if each quantum costs a constant capacity of c parameters, a size-N model can hold $m = N/c$ quanta:

$$L(N) - L_\infty \;\propto\; N^{-\alpha} \qquad \Longrightarrow \qquad \boxed{\alpha_N = \alpha}$$

— the model-size exponent *is* the data's Zipf exponent.

Exercise

Derive the other two laws yourself — the *data-limited* and *training-limited* exponents — and test the model against Kaplan's measured values. Full walkthrough on the Part III exercise sheet.

第 51 页：演算一条规律；另外两条留给你

尾和满足

$$L_m - L_\infty = \Delta \sum_{k>m} p_k \propto \int_m^\infty k^{-(\alpha+1)}dk \propto m^{-\alpha}.$$

参数受限：如果每个量子需要固定的 c 个参数容量，则大小为 N 的模型可以容纳 $m = N/c$ 个量子：

$$L(N) - L_\infty \propto N^{-\alpha} \qquad \Longrightarrow \qquad \alpha_N = \alpha.$$

也就是说，模型大小指数就是数据的 Zipf 指数。

习题。自行导出另外两条规律——数据受限和训练受限指数——并用 Kaplan 测量值检验模型。完整推导见第三部分习题纸。

References & further reading (Part III)

- J. Kaplan et al., *Scaling Laws for Neural Language Models* (2020); T. Henighan et al. (2020) for other modalities.
- J. Hoffmann et al., *Training Compute-Optimal Large Language Models* (2022) — Chinchilla; T. Porian et al. (2024) for the reconciliation with Kaplan et al.
- E. Michaud, E. Liu, U. Girit, and M. Tegmark, *The Quantization Model of Neural Scaling* (2023); blog companion at `ericjmichaud.com/quanta`.
- A. Maloney, D. Roberts, and J. Sully, *A Solvable Model of Neural Scaling Laws* (2022); B. Bordelon, A. Atanasov, and C. Pehlevan, *A Dynamical Model of Neural Scaling Laws* (2024).
- The long lecture notes, *Physics of Deep Learning*, final section, with the survey table and worked solutions.

第 52 页：参考文献与延伸阅读（第三部分）

- J. Kaplan 等，*Scaling Laws for Neural Language Models*（2020）；其他模态见 T. Henighan 等（2020）。
- J. Hoffmann 等，*Training Compute-Optimal Large Language Models*（2022），即 Chinchilla；与 Kaplan 等结果的协调见 T. Porian 等（2024）。
- E. Michaud、E. Liu、U. Girit 和 M. Tegmark，*The Quantization Model of Neural Scaling*（2023）；配套博客 `https://ericjmichaud.com/quanta`。
- A. Maloney、D. Roberts 和 J. Sully，*A Solvable Model of Neural Scaling Laws*（2022）；B. Bordelon、A. Atanasov 和 C. Pehlevan，*A Dynamical Model of Neural Scaling Laws*（2024）。
- 长讲义《深度学习的物理学》最后一节，包含综览表和详细解答。

译注：本地长讲义为 nosol 版本，不含实际习题解答；此处照录幻灯原文。